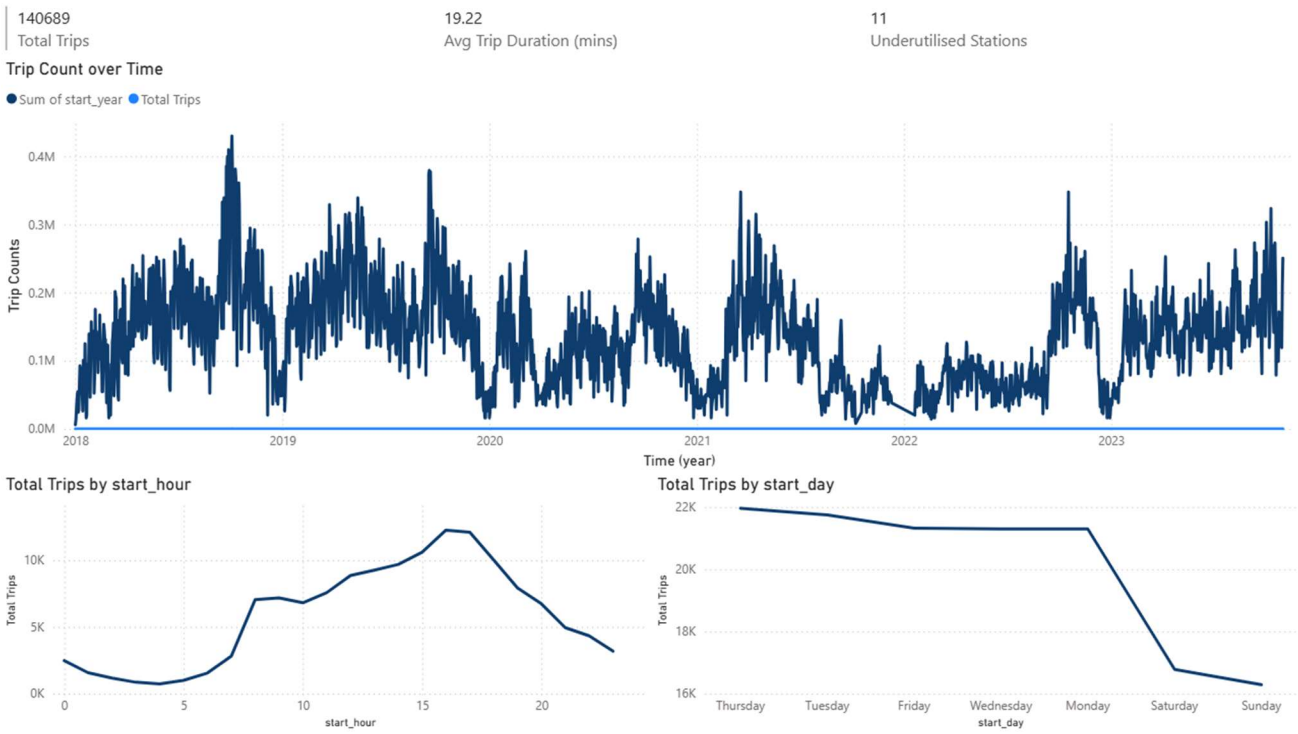
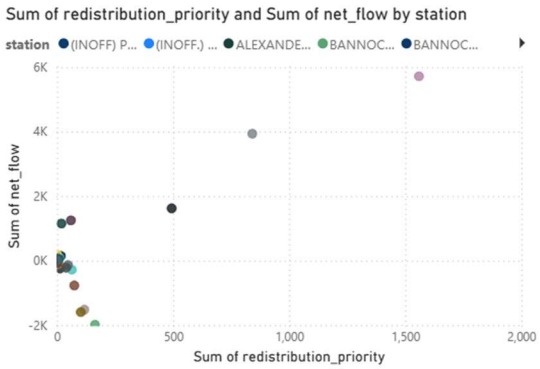
Optimising Nextbike Utilisation in Stirling

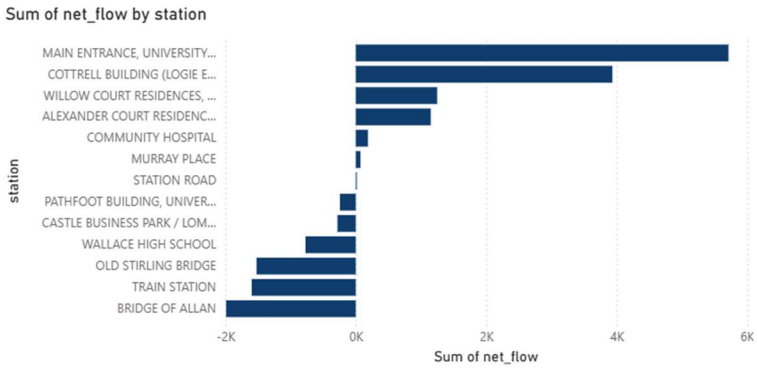
Transport accounts for over 37% of Scotland's greenhouse gas emissions, making sustainable transport central to achieving net-zero targets by 2045. In response, Stirling Council invested approximately £11 million in cycling infrastructure. However, infrastructure alone does not guarantee behavioural change. This project evaluates the utilisation of Stirling's Nextbike scheme and identifies how operational and behavioural interventions can increase its sustainability impact.

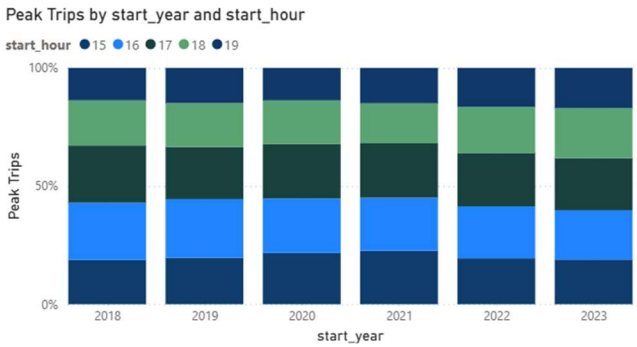


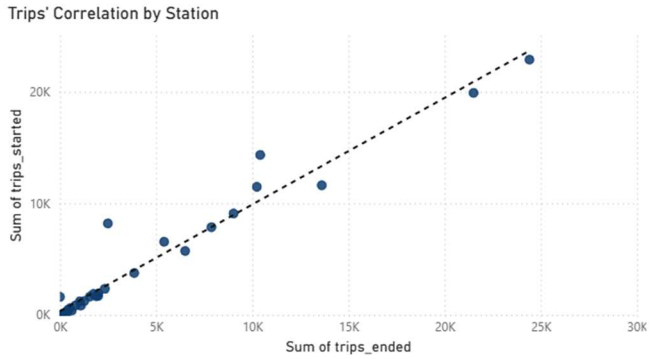
Using 2018–2023 Nextbike trip data (140,689 trips; average duration ≈19.7 minutes), our analysis reveals persistent underutilisation. Despite a fleet of 160 bikes, usage averages only ~78 trips per day. Temporal analysis shows strong weekday dependence and a clear afternoon peak (3–5pm). Spatial flow analysis highlights systematic imbalances: several city and transport-hub stations consistently receive more returns than rentals, while university-located stations experience sustained shortages. This indicates inefficient redistribution is the main issue rather than insufficient demand.

To address this bottleneck, we propose a pre-evening redistribution trial. Between 14:00–15:00, 20–30 bikes in total would be reallocated from surplus stations to the highest-deficit university locations. This targeted top-up aims to ensure bike availability ahead of the evening rush, when unmet demand is highest.









To complement staff-led redistribution, we apply the EAST behavioural framework through return-based incentives, rewarding users for returning bikes to priority university stations during the afternoon window (e.g. discounts or loyalty points). Light-touch in-app enhancements, such as CO_2-saved feedback, streaks, and targeted prompts, could further support behaviour change. In parallel, discussions with Active Stirling and the University are ongoing to pilot Freshers' Week awareness activities, supporting early habit formation. Broader demand-focused incentives are reserved for weekends and winter months, when usage is lowest.

Overall, the project primarily strengthens environmental sustainability by increasing active travel uptake and maximising the carbon-reduction returns of existing infrastructure investment, with additional social and economic co-benefits.

Reference

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